Unit 6 Seminar: K-Means Clustering Tutorial

Introduction

I ran the supplied k-means demo, corrected in five places, and all three tasks, using labels only after clustering and testing whether scaling and missing data change the answer.

Data preparation

Géron (2025) advises scaling before k-means. Euclidean distance is driven by how much each measurement varies: proline holds 99.77% of the raw wine variance, so I standardised the wine data. For iris, I used the raw measurements as a baseline and checked z-score and range scaling. For weatherAUS, I removed the Yes/No rain columns and 25,079 incomplete rows, leaving 120,381 (82.8%), then standardised.

Findings

At k = 3, raw iris clusters matched 134 of 150 flowers (Figure 1; 89.3%; adjusted Rand index (ARI) 0.730; Hubert and Arabie, 1985, cited in Arabie and Hubert, 1992). Z-scores, although Géron (2025) says scaling generally helps, lowered this to 125 (ARI 0.620). Z-scored wine matched 172 of 178 (96.6%; ARI 0.897), against 70.2% raw. Dividing by the range, which Milligan and Cooper (1988, cited in Arabie and Hubert, 1992) found better, came close to the best on both (88.7% and 94.9%). The silhouette score (Géron, 2025) disagreed with the labels twice: among k = 2 to 8 it favoured k = 2 for raw iris, despite three species, and raw wine scored higher than z-scored (0.571 against 0.285) while matching far worse. Weather silhouettes stayed low, 0.16 to 0.24. Filling gaps with medians instead of dropping rows gave almost the same k = 3 clusters (ARI 0.968; Table 3).

Implications and conclusion

Internal scores and tidy plots do not prove that groups exist; a two-dimensional plot cannot show separation in twelve dimensions. Without location or date, the weather clusters cannot be interpreted by station or season. In the demo, filling unknown default status with 0 assumes no default, which would need justifying before any customer decision. The UCI datasets' CC BY 4.0 licence requires the attribution kept here. I would record scaling and missing-data choices for whoever acts on the clusters.

Iris and wine evidence

Dataset, k = 3	Matched labels	ARI	Silhouette	Highest silhouette, k = 2 to 8
Iris, raw	134 of 150 (89.3%)	0.730	0.553	k = 2: 0.681
Iris, z-score	125 of 150 (83.3%)	0.620	0.459	
Iris, range	133 of 150 (88.7%)	0.716	0.504	
Wine, raw	125 of 178 (70.2%)	0.371	0.571	
Wine, z-score	172 of 178 (96.6%)	0.897	0.285	k = 3: 0.285
Wine, range	169 of 178 (94.9%)	0.854	0.301	

Table 1: Agreement with the labels after pairing each cluster with the label it overlaps most, one to one. This is agreement after fitting, not prediction accuracy. Silhouette is measured in the representation each model was fitted on, so raw and standardised scores describe different geometries. Range scaling divides each measurement by its range. The k scan was run on raw iris and z-scored wine.

Iris species	Cluster 0	Cluster 1	Cluster 2
setosa	0	50	0
versicolor	48	0	2
virginica	14	0	36

Wine cultivar	Cluster 0	Cluster 1	Cluster 2
1	0	0	59
2	65	3	3
3	0	48	0

Table 2: Labels against k = 3 clusters, raw iris and z-scored wine. Cluster numbers are arbitrary.

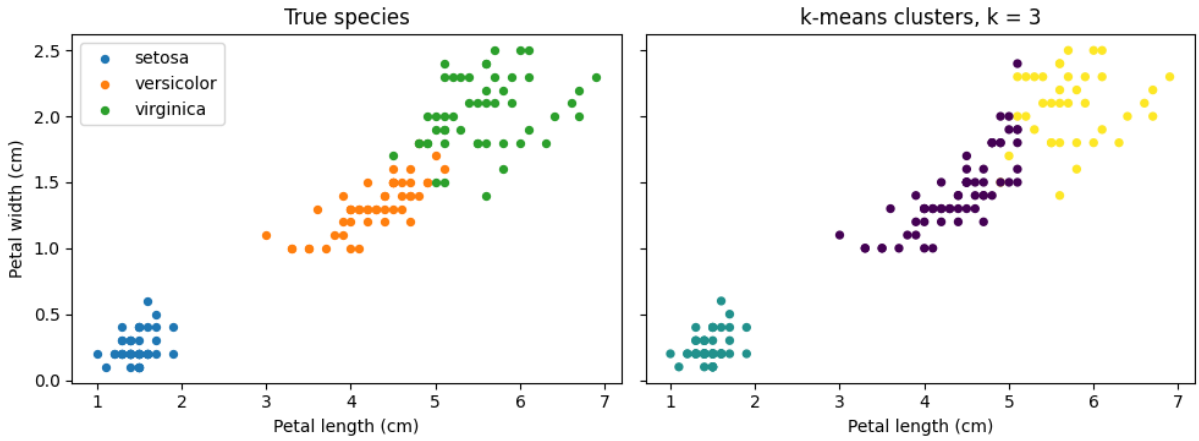


Figure 1: Iris petal measurements by true species (left) and by k = 3 cluster (right). Setosa separates cleanly; versicolor and virginica overlap.

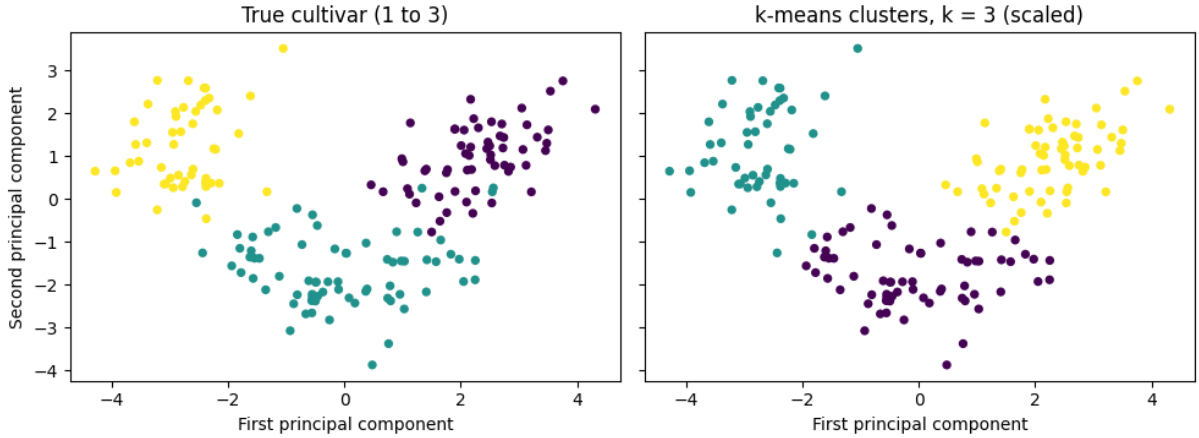


Figure 2: Wine on its first two principal components (55.4% of the standardised variation), by cultivar (left) and by standardised k = 3 cluster (right). Clustering used all 13 measurements.

Weather evidence

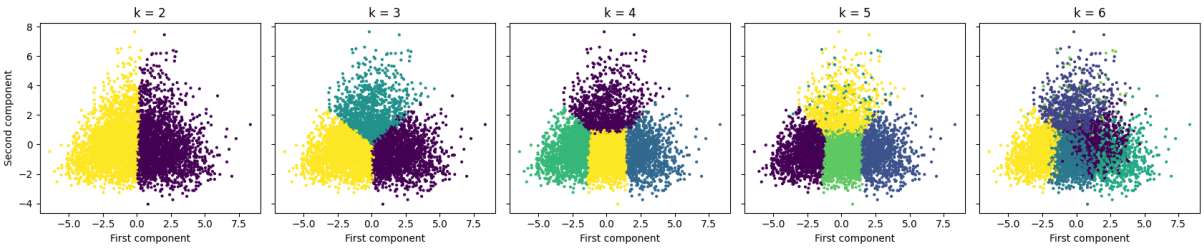


Figure 3: Weather clusters for k = 2 to 6, shown on the first two principal components (60.3% of the variation), for a random 5,000 of the 120,381 rows.

k	2	3	4	5	6
Incomplete rows dropped (120,381 rows)	0.237	0.229	0.160	0.172	0.168
Gaps filled with medians (145,207 rows)	0.226	0.218	0.154	0.165	0.156

Table 3: Weather silhouette scores on a random 10,000-row sample. The second version keeps every row with at least one measurement and fills 60,693 gaps with column medians; its k = 3 clusters agree with the first version on the 120,381 shared rows (ARI 0.968). Neither version establishes a correct k.

k = 2 cluster	Rows	Max temp (°C)	Humidity 3pm (%)	Pressure 3pm (hPa)	Rain today	Rain tomorrow
0	57,419	28.9	44.5	1011.2	17.6%	20.2%
1	62,962	18.5	56.7	1018.9	26.3%	23.5%

Table 4: Average measurements and share of rainy days in the two weather clusters. The rain columns were not used for clustering; blank rain values are left out of the shares.

Supplied demo

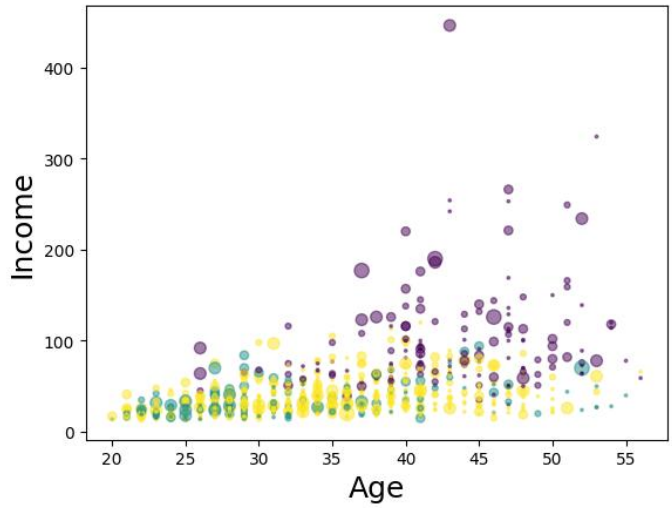


Figure 4: The corrected demo: 850 customers by age and income, coloured by k = 3 cluster fitted on the scaled data; marker size shows education level.

References

Aeberhard, S. and Forina, M. (1992) *Wine* [Dataset]. UCI Machine Learning Repository. Available at: https://doi.org/10.24432/C5PC7J

Arabie, P. and Hubert, L.J. (1992) 'Combinatorial data analysis', *Annual Review of Psychology*, 43, pp. 169–203. Available at: https://doi.org/10.1146/annurev.ps.43.020192.001125

Fisher, R.A. (1936) *Iris* [Dataset]. UCI Machine Learning Repository. Available at: https://doi.org/10.24432/C56C76

Géron, A. (2025) *Hands-On Machine Learning with Scikit-Learn and PyTorch*. Santa Rosa: O'Reilly.

IBM (no date) *Customer Segmentation Teaching Dataset* [Dataset]. Available at: https://s3-api.us-geo.objectstorage.softlayer.net/cf-courses-data/CognitiveClass/ML0101ENv3/labs/Cust_Segmentation.csv (Accessed: 31 October 2026).

Young, J. (2020) *Rain in Australia* [Dataset]. Kaggle. Observations from the Australian Bureau of Meteorology. Available at: https://www.kaggle.com/datasets/jsphyg/weather-dataset-rattle-package (Accessed: 31 October 2026).

Supplied materials: University of Essex Unit 6 Seminar brief, K-Means demo notebook, Unit06 iris.csv, Unit06 wine.csv and Unit06 weatherAUS.csv.

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